

RECOMMENDATION SYSTEMS FOR PERSONALIZED CONTENT DISCOVERY

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WHAT ARE WE BUILDING?

Context:

- Every day there are billions of people using Recommendation Systems (Netflix, YouTube, Spotify) to receive recommendations for what to watch/listen to/do.
- The Netflix Prize was the first major machine learning competition to make a major advancement in how movies are recommended to customers.
- The dataset contains 100 million+ ratings by 480,189 users of 17,770 different movies.

Our Goal:

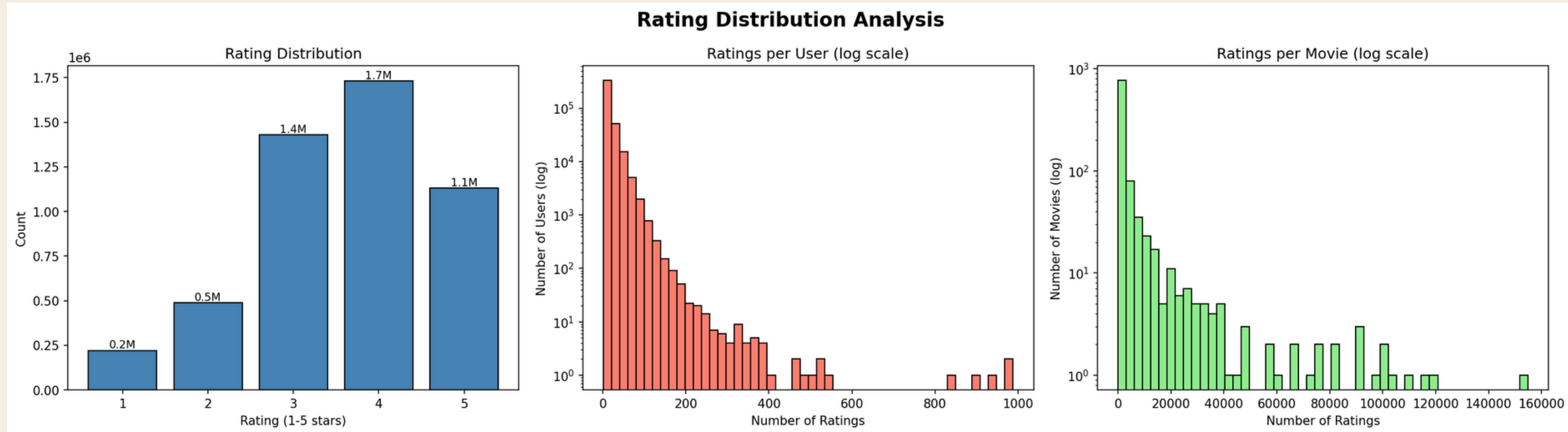
- Learn user preferences from historical ratings
- Predict ratings for movies users haven't seen
- Generate personalized Top-10 movie recommendations
- Evaluate using RMSE and MAP@10

DATASET AT A GLANCE

5,000,000 ratings loaded	404,478 unique users
996 movies in subset	17,770 movies in full catalog
1–5 star ratings	1999–2005
98.76% matrix sparsity	Source: combined_data_1.txt

- 98.76% of the user-movie interaction matrix is empty; therefore, each user has rated less than 2% of the movies available (in other words there is little "data" to train the recommendation system on). This fact makes implementing effective recommendation systems both difficult and necessary.
- The raw dataset is stored in a file-like format, where the MovieID occurs on its own line, and is followed by the UserID, Rating and Date (in one line records). A custom parser was built to read this file-like structure.

WHAT THE DATA TELLS US



- **Positivity Bias:** The combined rating of 4 stars and 5 stars accounts for 57% of all ratings made. Users are rating movies they have typically already identified as something they will likely enjoy.
- **Long-Tail Users:** The majority of users rate very few movies; therefore a small proportion of users (the "super users") account for the largest proportion of all ratings. This creates a "cold-start" problem for the majority of movies.
- **Popularity Skew:** A small number of movies receive 1,000's of ratings while most receive very few; therefore, the models must be designed to minimize recommending already-popular movies.

PROJECT APPROACH

SVD — Matrix Factorization:

Decomposing the rating matrix into user latent vectors and movie latent vectors, the rating prediction is determined using the dot product of the user latent vector and the movie latent vector plus user and movie bias, across 100 factors and 20 epochs.

NMF — Non-Negative MF:

This is similar to SVD, except for positive-only factors (more interpretable latent dimensions typically have some correlation with a theme/genre), across 15 factors and 20 epochs. Interpretable, but constrained.

SVD++ — Implicit Feedback:

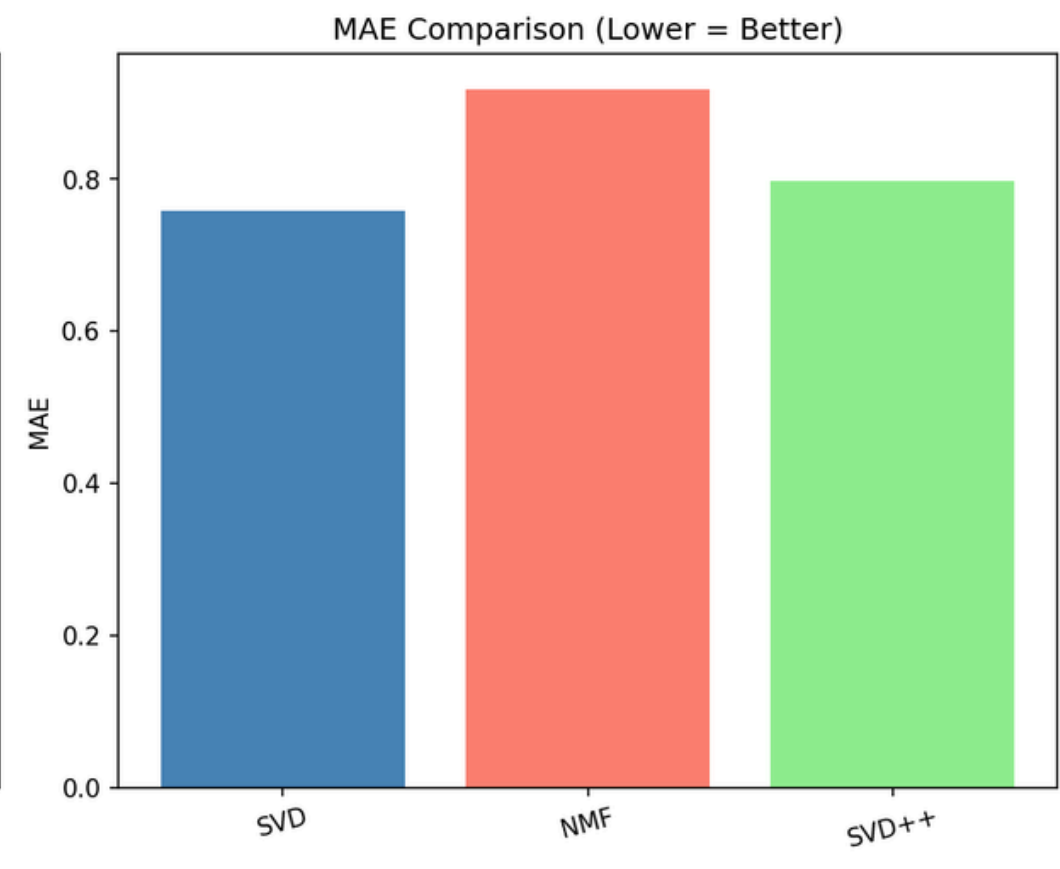
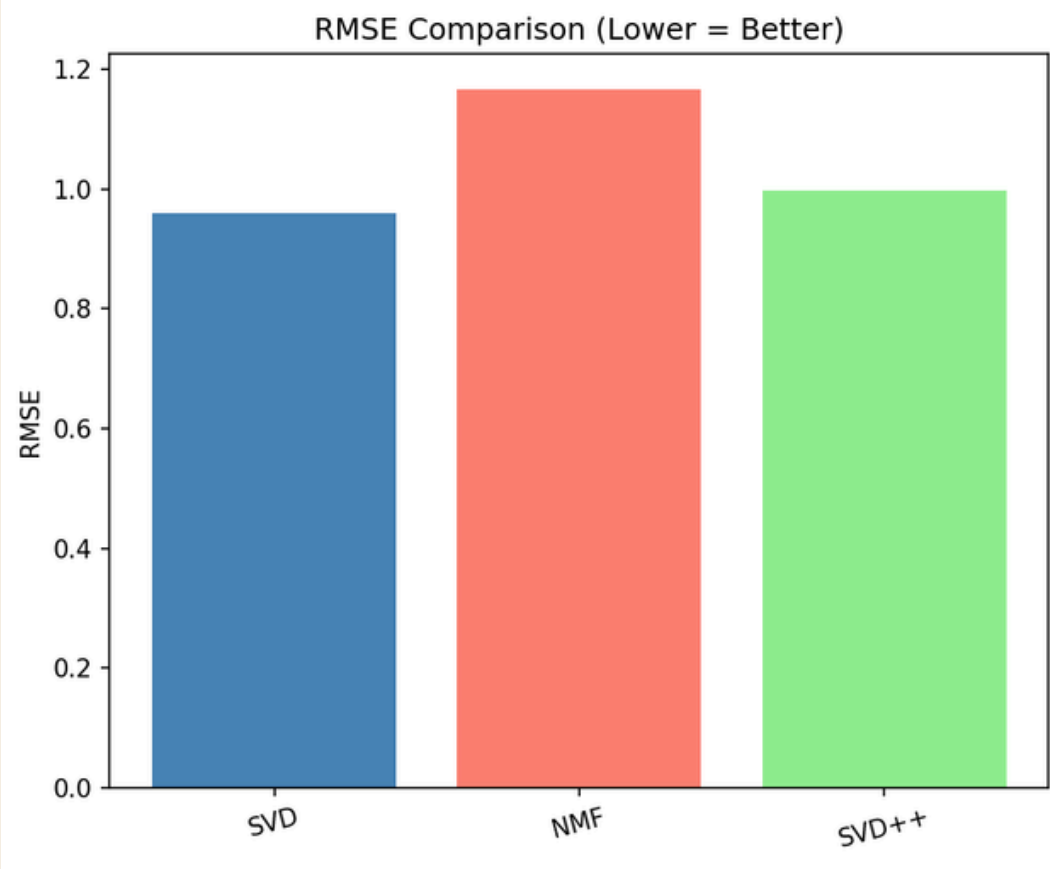
Augments the SVD framework by adding the user's movie rating (even without a score) to provide an implicit signal for what they are interested in (or likely interested in). 20 features, 20 epochs. Captures "hidden preferences".

NCF — Neural CF:

Deep learning approach composed by the following architecture : Fully connected layers (64-dim to 128-dim to 64-dim to 1-dim) with ReLU activation and dropout. Optimized using Adam for 5 epochs. Very expressive and using deep learning technology.

All models trained on 80/20 train-test split. SVD/NMF/SVD++ use Surprise library. NCF built in PyTorch.

MODEL PERFORMANCE COMPARISON



Model	RMSE	MAP@10	MAE
SVD	0.9594	0.6933	0.7580
NMF	1.1669	0.5819	0.9184
SVD+ +	0.9966	0.5862	0.7980
NCF	0.9941	N/A	-

- SVD wins on both metrics — best RMSE (0.9594) and best MAP@10 (0.6933)
- NCF is competitive — achieves 0.9941 RMSE trained from scratch in just 5 epochs, showing the promise of neural approaches

Relevance definition used for MAP@10: A movie is relevant if actual user rating ≥ 3.5

RECOMMENDATION EXAMPLES

Top 10 Recommendations for User 530789:

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- 1. Yojimbo (1961) – Predicted: 4.96 STAR
- 2. Ray (2004) – Predicted: 4.85 STAR
- 3. The Deer Hunter (1978) – Predicted: 4.83 STAR
- 4. Ghosts of Rwanda: Frontline (2004) – Predicted: 4.80 STAR
- 5. Sense and Sensibility (1981) – Predicted: 4.78 STAR
- 6. Futurama: Monster Robot Maniac Fun Collection (1999) – Predicted: 4.77 STAR
- 7. Lord of the Rings: The Return of the King: Extended Edition: Bonus Material (2003) – Predicted: 4.76 STAR
- 8. Seven Samurai (1954) – Predicted: 4.73 STAR
- 9. Xena: Warrior Princess: Season 3 (1995) – Predicted: 4.72 STAR
- 10. The Twilight Zone: Vol. 12 (1962) – Predicted: 4.68 STAR

Success Case:

Users 901650, 767537, 2641990, 1252974, 2234150 — 100% of Top-10 recommendations were relevant. These users had rich rating histories enabling accurate preference modeling.

Failure Case:

Users 17462 (38 ratings), 880344 (15 ratings), 163069 (20 ratings) — 0% of recommendations were relevant. Sparse histories make latent factor estimation unreliable → cold-start problem.

SUMMARY & WHAT'S NEXT

What We Achieved:

- Created four different recommendation models and compared them (SVD, NMF, SVD++, NCF)
- SVD reached an RMSE of 0.9594 and MAP@10 of 0.6933 — corresponding to about 7 of 10 recommendations being relevant for the user.
- NCF reached a competitive RMSE of 0.9941 trained from scratch using PyTorch.
- Examined both the cases of success and failure for recommendations; identified cold-starting as the main obstacle.

Future Improvements:

- Train on full dataset (all 4 files, ~100M ratings)
- Cold-start handling using movie metadata (genre, year)
- Hybrid model combining SVD + content-based filtering
- Extended NCF training (20+ epochs with GPU)
- Deploy as FastAPI web service for real-time recommendations
- Incorporate temporal dynamics — user tastes evolve over time

THANK YOU!